K-FLATNESS SATISFIES THE TWO-OUT-OF-THREE PROPERTY FOR SHORT EXACT SEQUENCES OF COMPLEXES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- K-flat complex, defined through the vanishing of the cohomology of the total tensor product with any acyclic complex. definition at source lines 14185-14191 of the same chapter. Supplies the property that is being proved and the property assumed for two of the three complexes.
- Flatness of a module implies that tensoring with it preserves injectivity, equivalently the vanishing of Tor. the author cites Algebra, Lemma algebra-lemma-flat-tor-zero (tag 00HL) at proof line 14272. Gives the degreewise exactness of the tensored sequence.
- In a short exact sequence of complexes, if two out of three are acyclic then so is the third. used at proof lines 14278-14281. Transfers the vanishing of cohomology from two complexes to the third.

Source reference. Original lemma 0BYH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15015–15015. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 647–647. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $0 \to K_1^\bullet \to K_2^\bullet \to K_3^\bullet \to 0$ be a short exact sequence of complexes. If K_3^n is flat for all $n \in \mathbf{Z}$ and two out of three of $K_i^\bullet$ are K-flat, so is the third.*

Proof. Let $L^\bullet$ be a complex of R-modules. Then

$$0 \to \mathrm{Tot}(L^\bullet \otimes_R K_1^\bullet) \to \mathrm{Tot}(L^\bullet \otimes_R K_2^\bullet) \to \mathrm{Tot}(L^\bullet \otimes_R K_3^\bullet) \to 0$$

is a short exact sequence of complexes. Namely, for each n, m the sequence of modules $0 \to L^n \otimes_R K_1^m \to L^n \otimes_R K_2^m \to L^n \otimes_R K_3^m \to 0$ is exact by Algebra, Lemma 00HL and the sequence of complexes is a direct sum of these. Thus the lemma follows from this and the fact that in a short exact sequence of complexes if two out of three are acyclic, so is the third. □